Fiber Optics

(Unit – 3)

Introduction

- An optical fiber is a dielectric waveguide which guide the light wave by total internal reflection.
- Advantages of optical fiber include:
  - Greater bandwidth than copper
  - Lower loss
  - Immunity to **crosstalk (Immune to EMI)**
  - No electrical hazard
  - Data transmission is more secure

Total Internal Reflection (TIR)

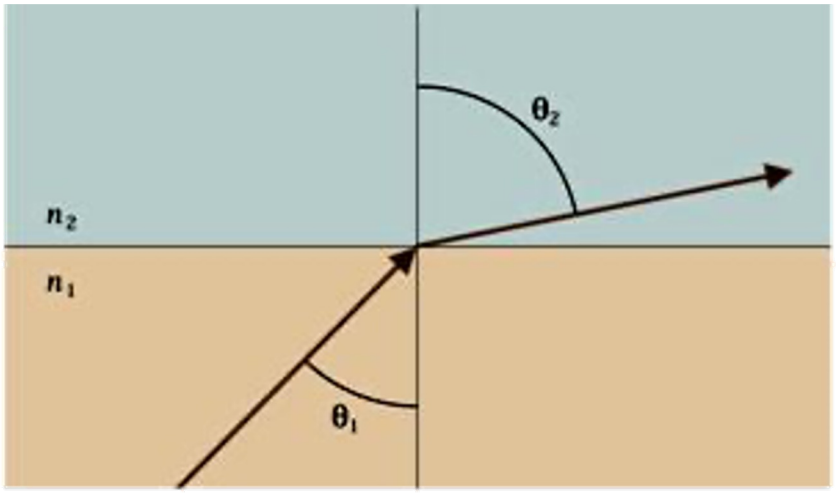
Optical fibers work on the principle of **total internal reflection**

Conditions for TIR

1. Incident angle ≥ Critical angle

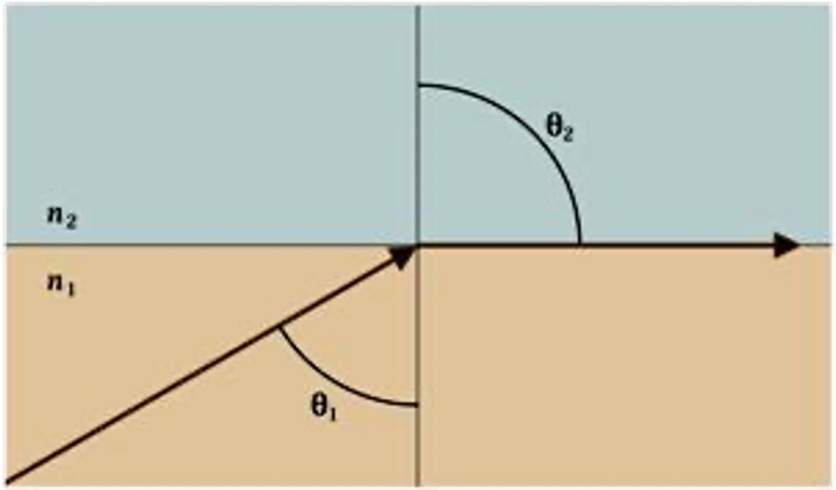
2. Light must incident from denser to rarer medium

Refraction & Total Internal Reflection

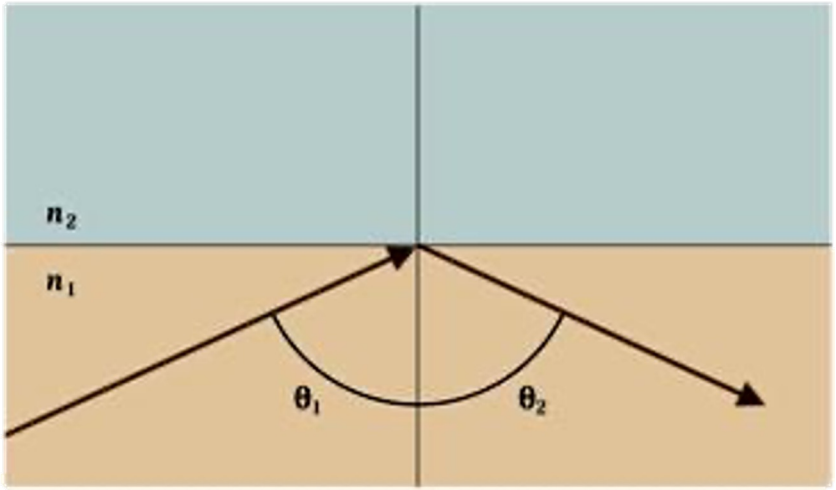


(a) Angle of incidence less than critical angle

$n_1 > n_2$



(b) Angle of incidence equal to critical angle



(c) Angle of incidence greater than critical angle

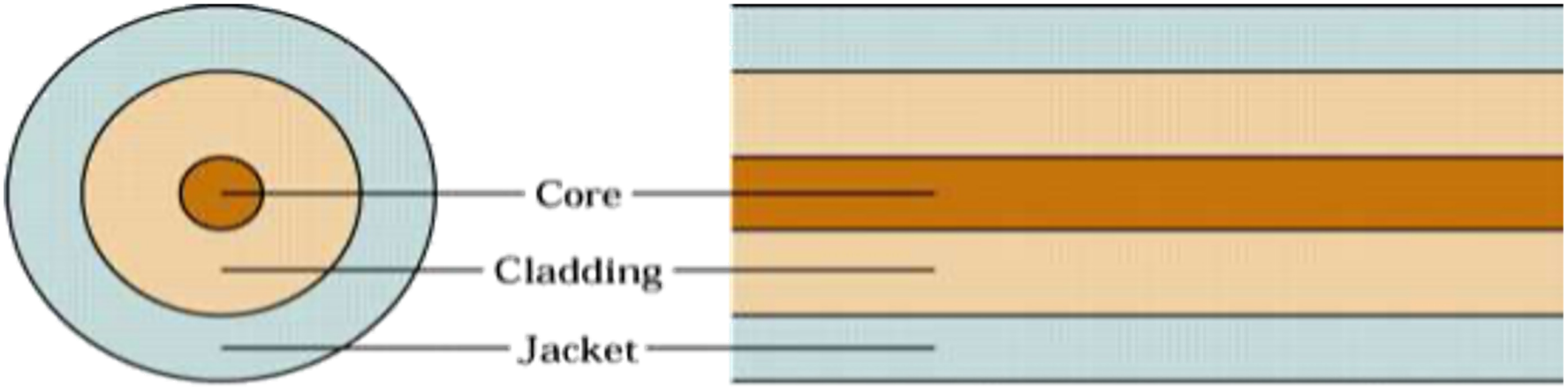
The **angle of refraction** at the interface between two media is governed by Snell's law:

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

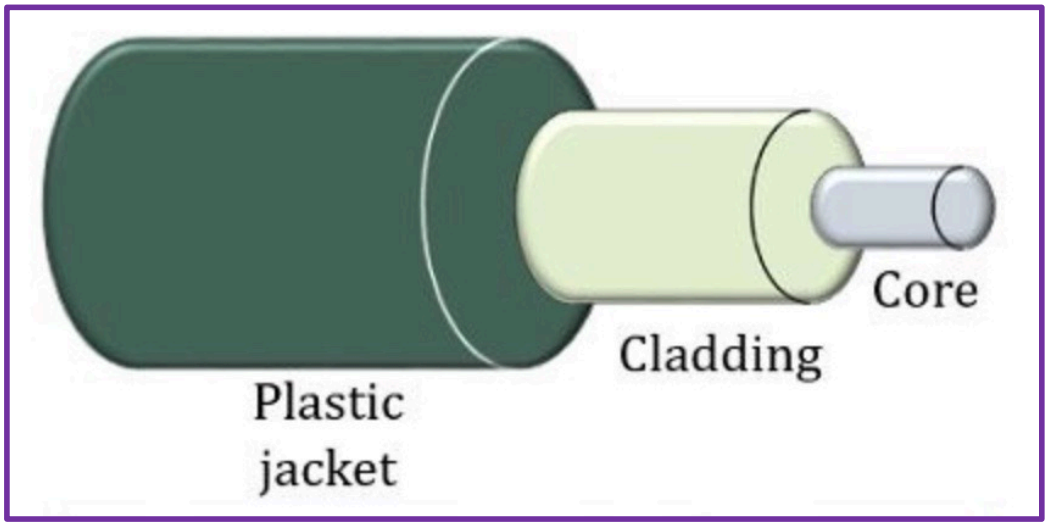
When $\theta_1 = \theta_c$ (critical angle) then $\theta_2 = 90^0$

$$\sin \theta_c = \frac{n_2}{n_1}$$

Optical Fiber



(a) Fiber cross section



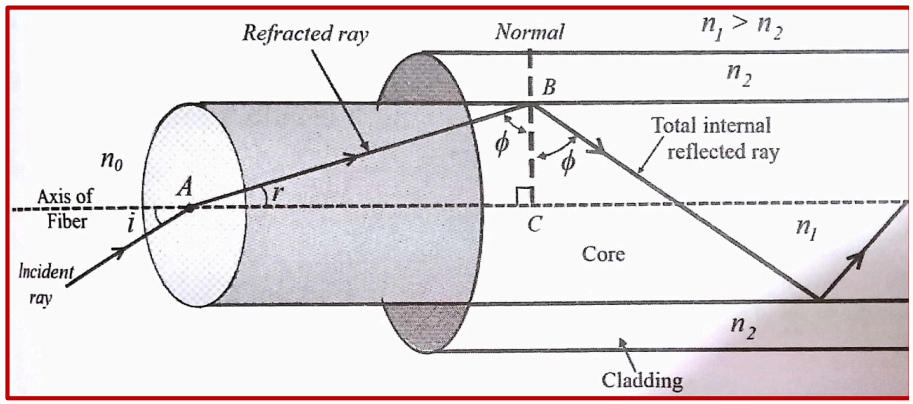
Acceptance angle

Applying Snell's law at point A

$$\frac{\sin\theta_i}{\sin\theta_r} = \frac{n_1}{n_0} \quad \text{or} \quad \sin\theta_i = \frac{n_1}{n_0}\sin\theta_r$$

But in right angled triangle ABC $\sin\theta_r = \sin(90^\circ - \emptyset) = \cos\emptyset$

$$\therefore \quad \sin\theta_i = \frac{n_1}{n_0}\cos\emptyset$$



When $\phi = \phi_c$ then $\theta_i = \theta_{max}$

Thus, $$\sin\theta_{max} = \frac{n_1}{n_0}\cos\emptyset_c$$

Now, applying Snell`s law at core-cladding interface,

$$\frac{\sin\emptyset_c}{\sin 90^\circ} = \frac{n_2}{n_1} \quad \text{or} \quad \sin\emptyset_c = \frac{n_2}{n_1}$$

or $$\sqrt{(1 - \cos^2\emptyset_c)} = \frac{n_2}{n_1}$$

or $$1 - \cos^2\emptyset_c = \frac{{n_2}^2}{{n_1}^2}$$

$$\therefore \qquad \cos \emptyset_c = \frac{\sqrt{n_1{}^2 - n_2{}^2}}{n_1}$$

Hence, $$\sin \theta_{max} = \frac{\sqrt{n_1{}^2 - n_2{}^2}}{n_0}$$

Therefore acceptance angle

$$\boxed{\theta_{max} = \sin^{-1} \frac{\sqrt{n_1{}^2 - n_2{}^2}}{n_0}}$$

For air $n_0 = 1$

$$\boxed{\theta_{max} = \sin^{-1} \sqrt{n_1{}^2 - n_2{}^2}}$$

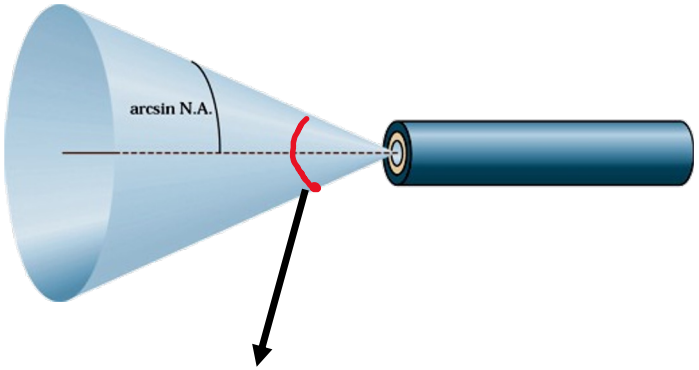
The maximum angle (θ_{max}) that a light wave can have relative to axis of the fiber for its propagation thorough fiber.

Numerical Aperture (NA)

- The **numerical aperture** measure light gathering capacity of an optical fiber.
- The numerical aperture is given by the formula (in air):

$$N.A. = \sqrt{n_1^2 - n_2^2}$$

- NA of fiber is sine of **acceptance angle**.
- The **acceptance cone angle** is twice of the acceptance angle.



Acceptance cone angle

$\sin\theta_{max}$ ***is known as Numerical Aperture***

Fractional refractive index change

The ratio of the refractive index difference between the core and the cladding to the refractive index of core.

$$\Delta = \frac{(n_1 - n_2)}{n_1}$$

$$N.A = \sqrt{n_1^2 - n_2^2}$$

$$N.A = \sqrt{(n_1 + n_2)(n_1 - n_2)}$$

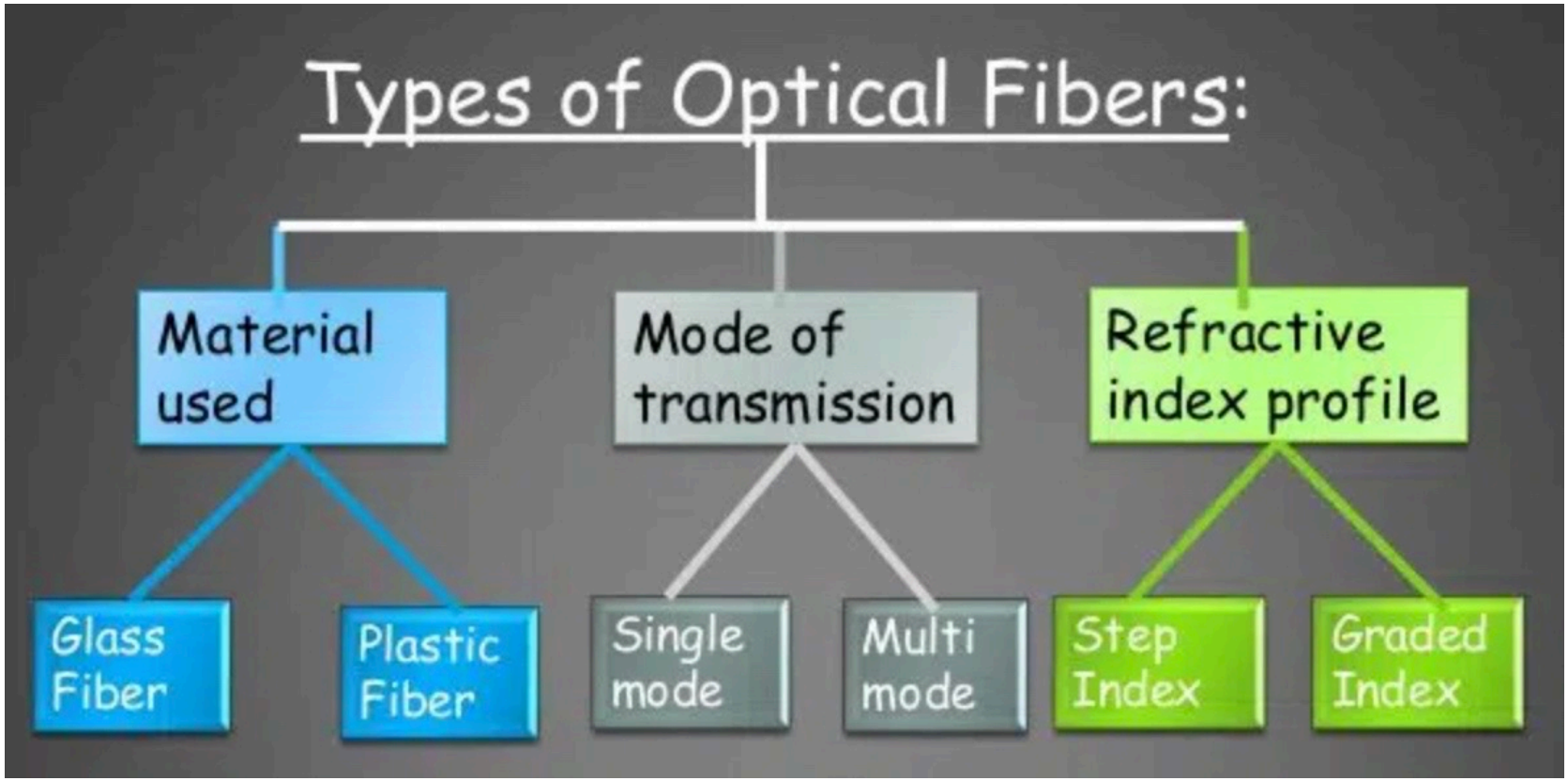
$$N.A = \sqrt{(n_1 + n_2)n_1\Delta}$$

In optical fiber difference between n_1 and n_2 is very small, $n_1 \approx n_2$

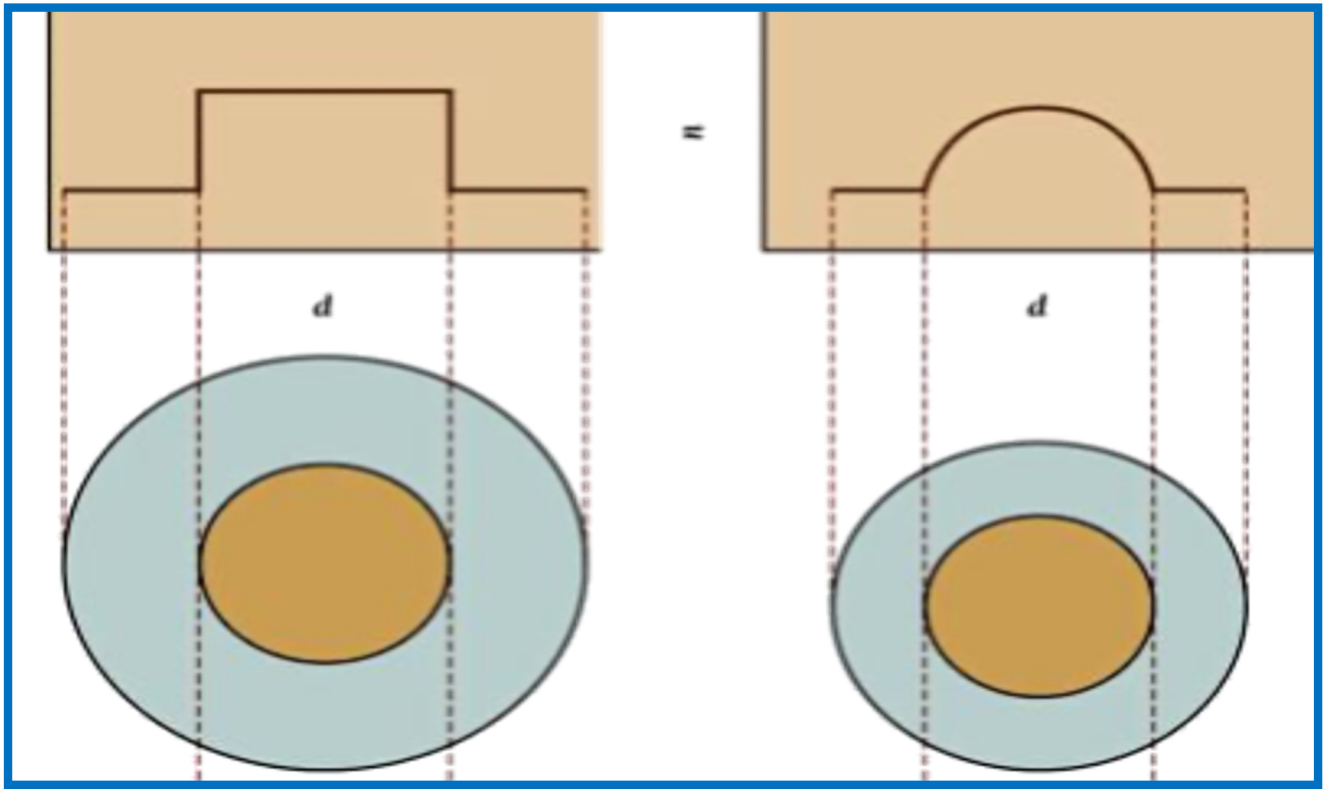
$$N.A = \sqrt{2n_1^2\Delta}$$

$$N.A = n_1\sqrt{2\Delta}$$

Types of Fiber



Refractive index profile



Step index fiber

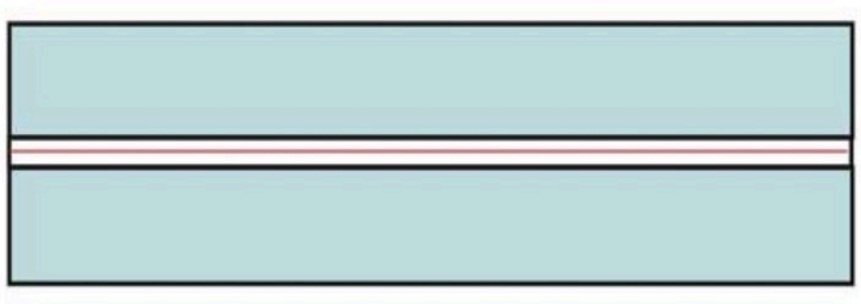
Graded index fiber

Refractive index of core is constant

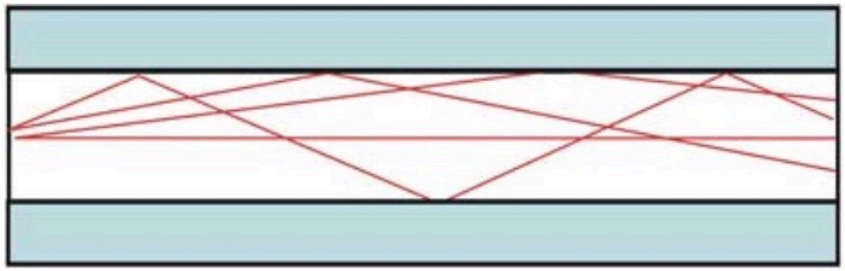
Refractive index of core gradually decreases away from the center of the core

Modes

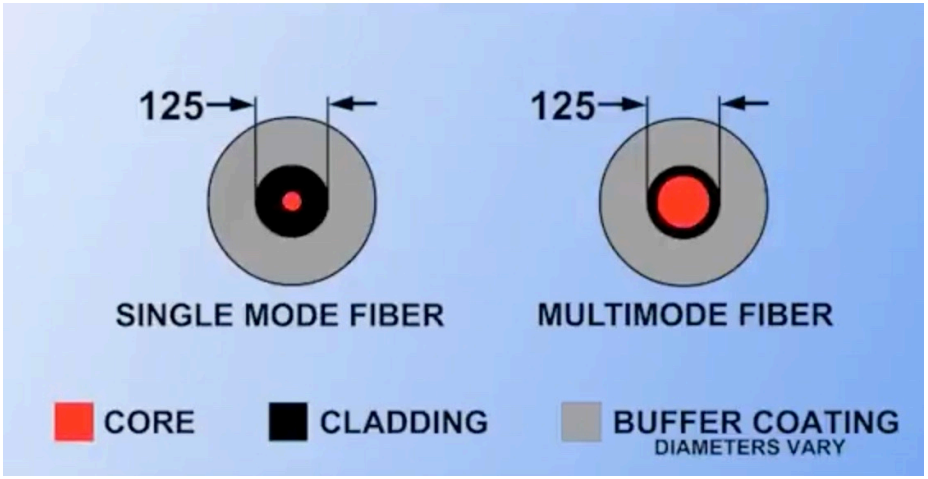
Electric field distributions (path available to light) in fiber

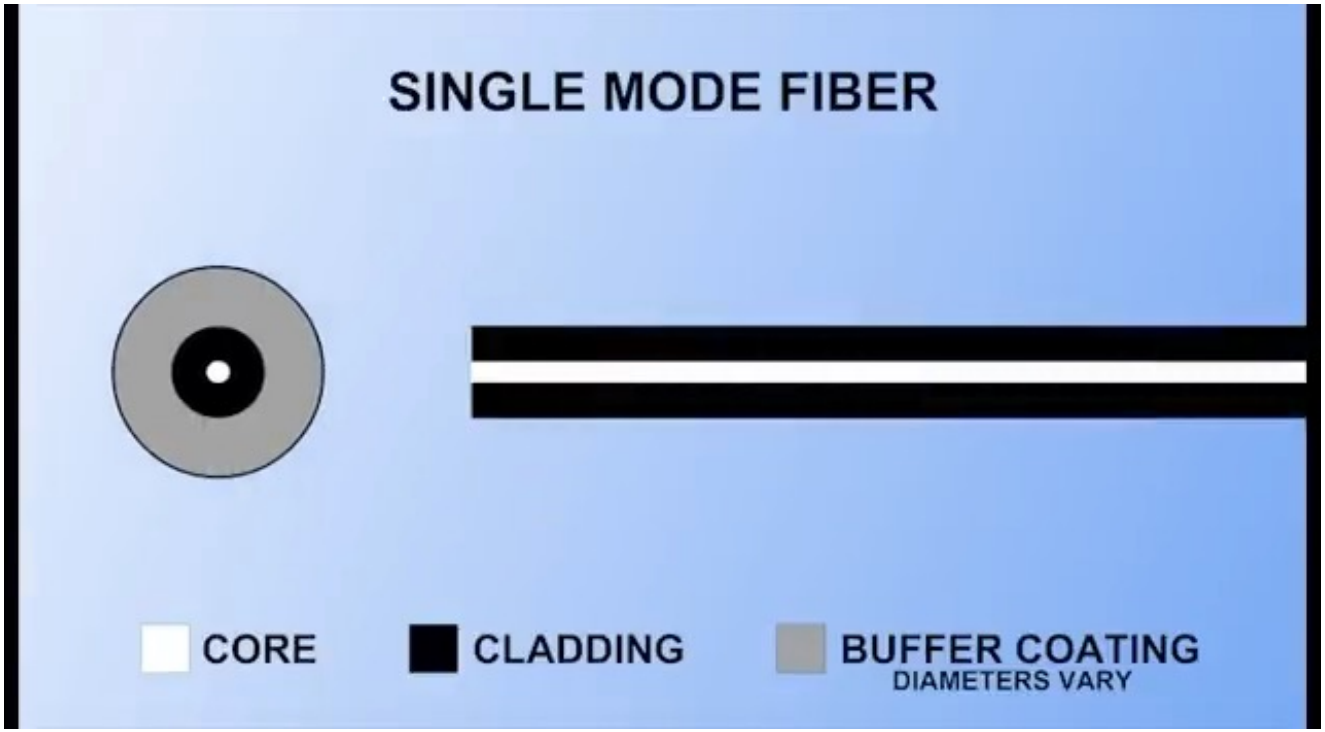


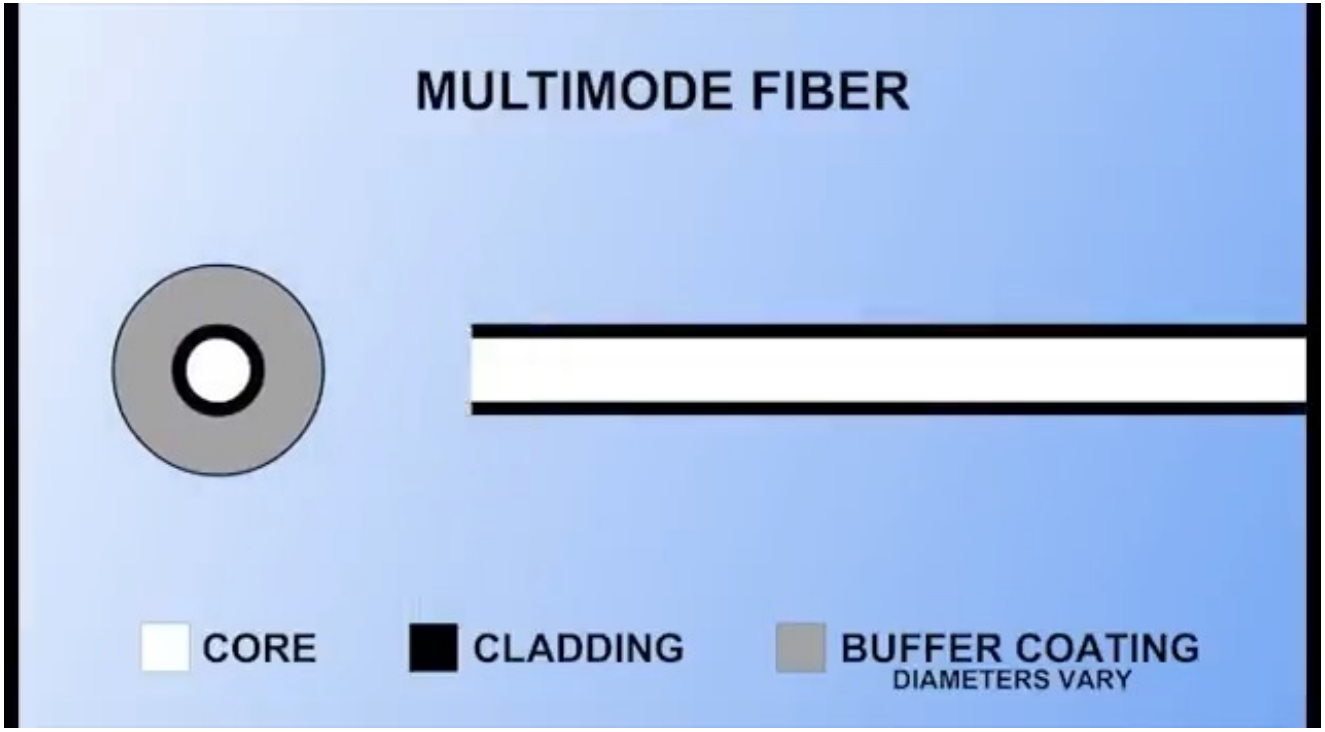
Single Mode / Mono-mode fiber



Multimode fiber




SINGLE MODE FIBER
CORE
CLADDING
BUFFER COATING
DIAMETERS VARY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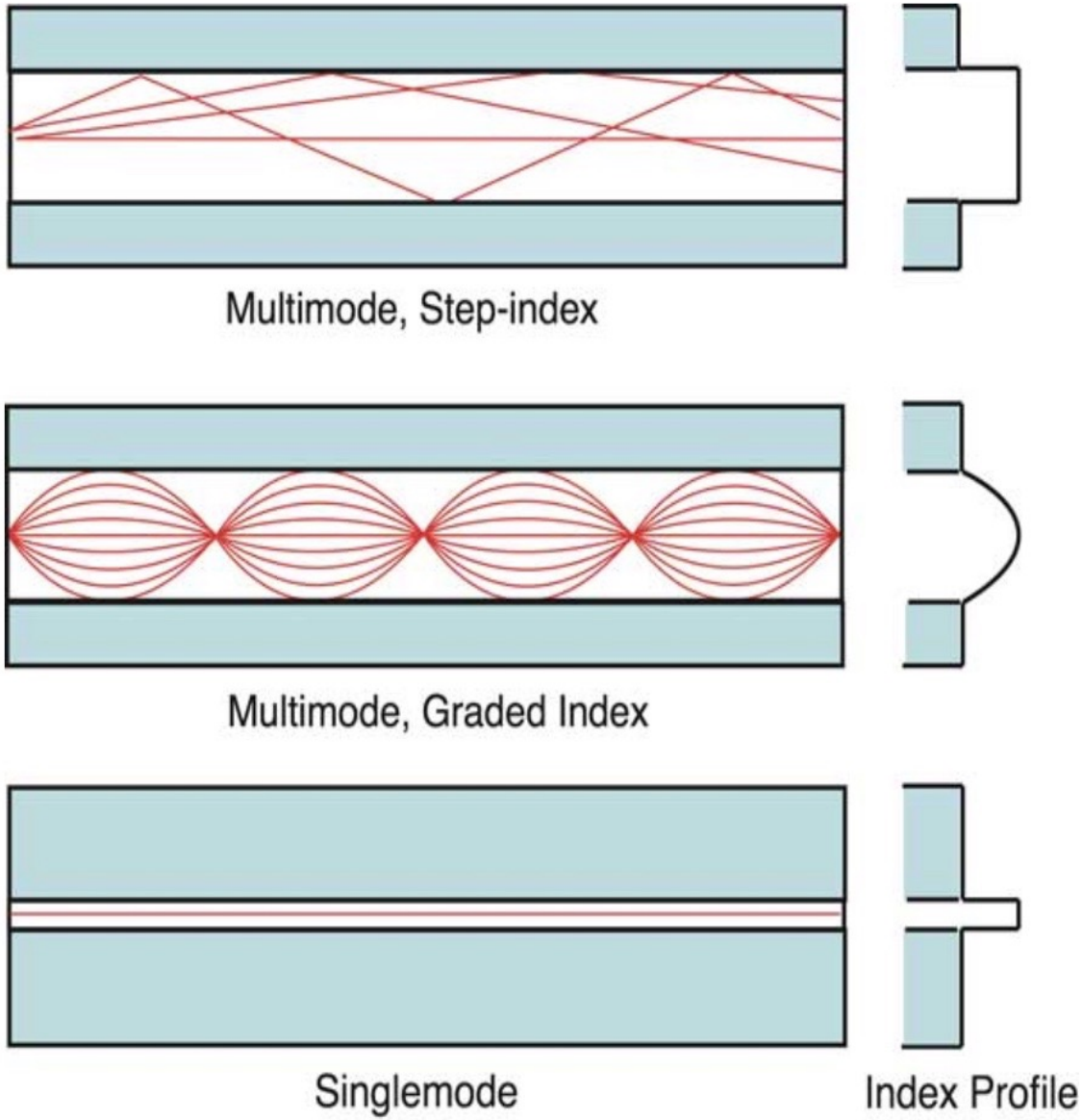
Multimode, Step-index
Multimode, Graded Index
Singlemode
Index Profile

Comparison

SINGLE MODE FIBER	MULTIMODE FIBER
In single mode fiber only one mode can propagate through the fiber	In multimode it allows a large number of paths or modes for the light rays travelling through it.
It has smaller core diameter and the difference between the refractive index of the core and cladding is very small.	It has larger core diameter and refractive index difference is larger than the single mode fiber.
Advantages: ~~No~~ dispersion(i.e. there is no degradation of signal during propagation)	**Disadvantages:** Dispersion is more due to degradation of signal owing to multimode.

less

Disadvantages: Launching of light and connecting of two fibers difficult. Installation (fabrication) is difficult as it is more costly	**Advantages:** Launching of light and also connecting of two fibers is easy. Fabrication is easy and the installation cost is low.

Single mode fiber	Multi mode fiber
Only one mode is transmitted through the fiber.	Many modes are transmitted through the fiber.
It has smaller core diameter Up to 10 μm	It has larger core diameter. ~ 50 μm, 62.5 μm
Difference between the refractive index of core and cladding is very small.	Difference between the refractive index of core and cladding is large.
NA is small. ~ 0.1	NA is large. ~ ≥ 0.2
High information carrying capacity.	Low information carrying capacity.

Comparison

STEP INDEX FIBER	GRADED INDEX FIBER
The refractive index of the core is uniform throughout and undergoes on abrupt change at the core cladding boundary	The refractive index of the core is made to vary gradually such that it is maximum at the center of the core.
The diameter of the core is about 50-200μm in the case of multimode fiber and 10μm in the case of single mode fiber	The diameter of the core is about 50μm in the case of multimode fiber
The path of light propagation is ***zig- zag*** in manner	The path of light is ***helical*** in manner
Attenuation is more for multimode step index fiber but for single mode it is very less. *Explanation:* When a ray travels through the longer distances there will be some difference in reflected angles. Hence high angle rays arrive later than low angle rays causing dispersion resulting in distorted output.	***Attenuation is less.*** *Explanation:* Here the light rays travel with different velocity inn different paths because of their variation in their refractive indices. At the outer edge it travels faster than near the center. But almost all the rays reach the exit at the same time due to helical path. Thus, there is ~~no~~ less dispersion.